

EEG Seizure Detection & Subtype Classification

A plain-language summary of the five models built in this project — what each one does, its basic architecture, its results, and an honest comparison of pros and cons.

The underlying question every model here answers is the same: given a clip of brain-electrode (EEG) recording, **is this a seizure — and if so, what type?** Five things were built, each one building on lessons from the last. This document walks through them in the order they were built, then compares all five side by side.

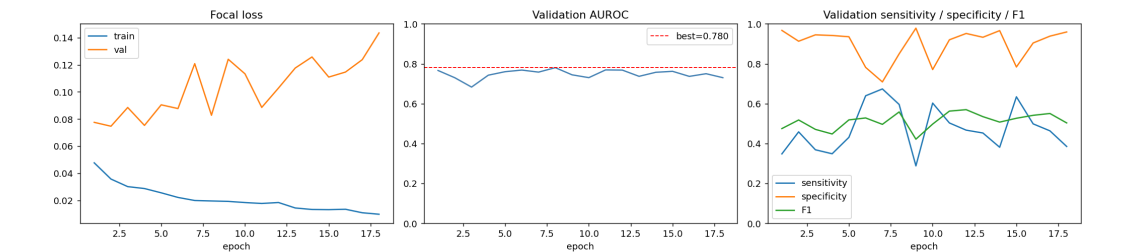
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Stage 1 — Seizure Detection (1D CNN)

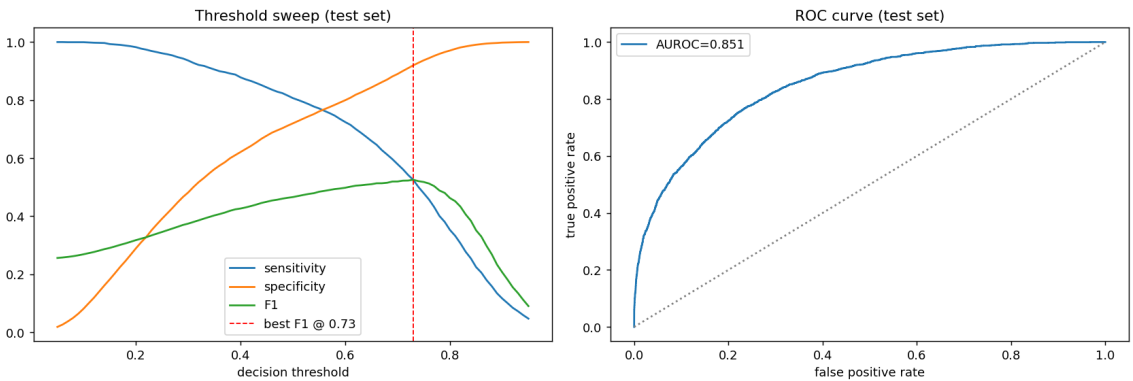
Detection only

What it does: looks at a 4-second window of 18-channel EEG and answers one yes/no question — is this a seizure? Trained on both datasets pooled together (CHB-MIT + Siena), so it sees the widest possible variety of patients.

Basic architecture: five stacked 1D-convolution blocks that shrink the signal down while learning increasingly abstract wave-shapes (18 → 32 → 64 → 128 → 256 channels), followed by average-pooling and a small decision layer that outputs one number: the probability the window is a seizure.



Training progress: loss falling, validation AUROC climbing (best epoch marked).



ROC curve and decision-threshold sweep on the held-out test patients.

Results (held-out test patients, never seen during training):

Metric	Value
Validation AUROC	0.780
Test AUROC	0.851
Best decision threshold	0.73
Sensitivity at that threshold	0.525
Specificity at that threshold	0.920
F1 score	0.526

Pros

- Simple, fast to train (~20 min)
- Solid AUROC (0.85) on unseen patients
- Uses both datasets — sees a wide variety of patients

Cons

- Window-level only — no smoothing across a recording
- No idea of seizure *type*, only yes/no
- Sensitivity/specificity is a tunable trade-off, not free lunch

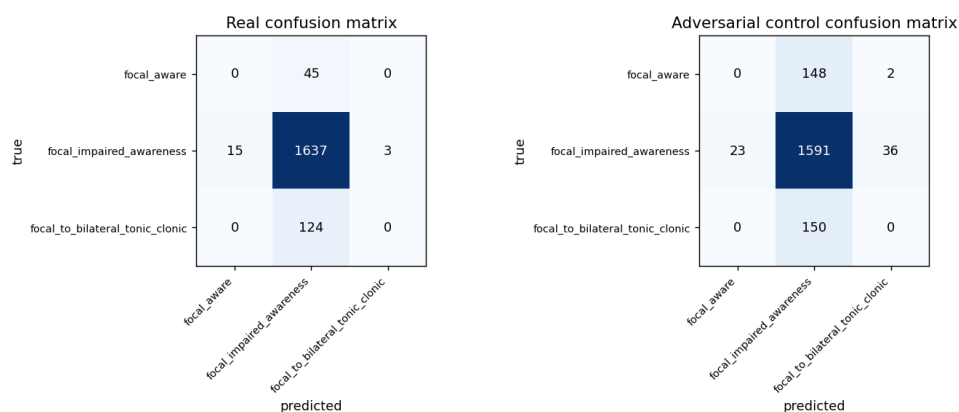
Stage 2 — Seizure Subtype Classification (1D CNN)

Subtype only

What it does: assumes a window is *already known* to be a seizure, and asks *which of 3 types* it is (per the ILAE classification): focal aware, focal impaired awareness, or focal → bilateral tonic-clonic. Only Siena has this label, so this model only ever sees Siena's 13 usable patients.

Basic architecture: the same 5-block convolutional design as Stage 1, just narrower (less capacity, since there's ~500× less data), ending in a 3-way decision layer instead of a yes/no one.

Why evaluation is unusual here: 13 patients, and 2 of the 3 types belong to exactly **one** patient each. A normal train/test split would let the model just memorize a patient's personal EEG signature instead of learning the seizure type — so every patient is held out in turn (leave-one-subject-out, or "LOSO"), and an **adversarial control** reruns the identical test on plain background (non-seizure) signal, relabeled with each patient's own type. If the control score is nearly as high as the real score, the model is just recognizing the patient, not the seizure.



Real seizure windows (left) vs. the adversarial control (right) — same axes, same protocol.

Results (aggregated across all 13 leave-one-patient-out folds):

	Accuracy	Macro-F1
Real seizure windows	89.7%	0.315
Adversarial control (patient-recognition floor)	81.6%	0.300
Real minus control (the number that matters)	+8.2 pts	+0.015

Pros

- Correctly identifies the majority class almost every time
- Rigorous methodology: LOSO + adversarial control catch a common false-confidence trap

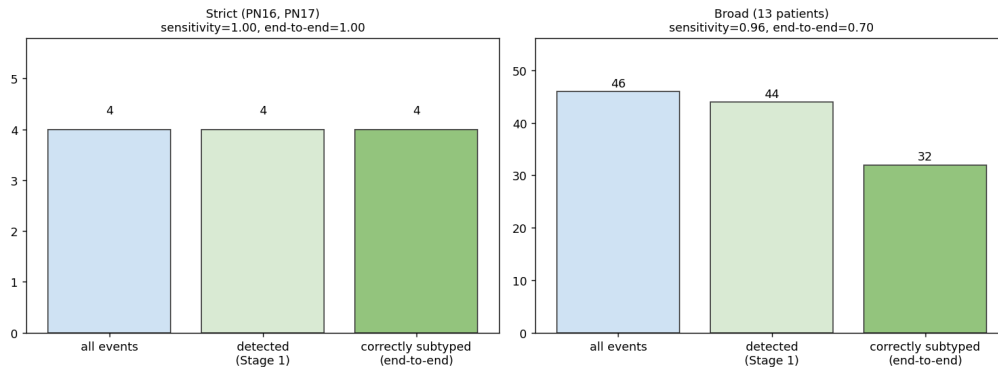
Cons

- 0% recall on **both** minority seizure types
- Real-vs-control margin (+0.015 macro-F1) is barely above the patient-recognition floor
- Ceiling is the data (1 patient per rare type), not something more training fixes

Cascade — Stage 1 + Stage 2 Wired Together

Detect, then classify

What it does: this isn't a new model — it's a test of the honest, real-world question: if you actually chain Stage 1 and Stage 2 together (detect first, only then classify), how well does the *whole pipeline* do? Stage 1 and Stage 2 above were each trained and judged in isolation; this wires them together after the fact, with no shared weights between them.



How many real seizure events survive each stage of the pipeline, strict vs. broad patient sets.

Two read-outs, reported separately on purpose: "Strict" uses only the 2 patients genuinely unseen by *both* stages (a small but fully honest test). "Broad" uses all 13 Stage-2 patients, most of whom Stage 1 already trained on — larger sample, but optimistic about Stage 1's numbers specifically.

	Detection sensitivity	Conditional subtype accuracy	True end-to-end accuracy	n events
Strict (2 patients)	100%	100%	100%	4
Broad (13 patients)	95.7%	74.4%	69.6%	46

Pros

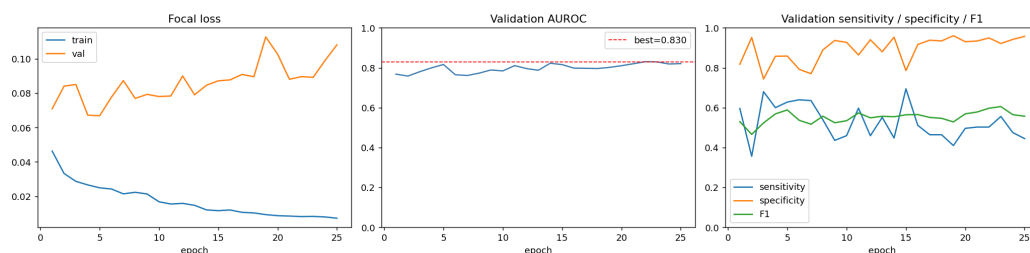
- Tests the actual deployed pipeline, not two models in a vacuum
- Honest strict-vs-broad reporting instead of one flattering number

Cons

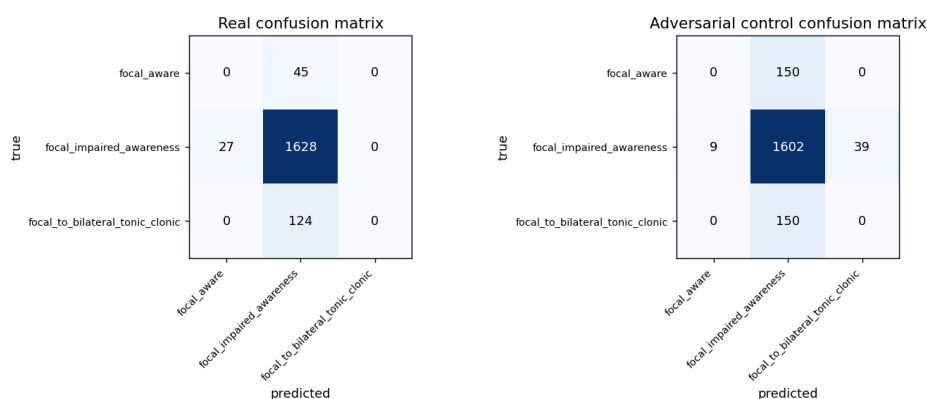
- Strict result is only 4 events — a real number, but not a statistically solid one
- Broad result is optimistic about Stage 1 (most patients were in its own training set)
- Stage 2 has no "not a seizure" option — a Stage 1 false alarm still gets a confident (wrong) subtype

What it does: replaces the two separate CNNs above with **one shared model**. A single backbone (convolution layers + a bidirectional LSTM + an attention layer) is trained once on detection, using both datasets pooled together. That trained backbone is then literally **copied** into the subtype classifier — not started from random weights — and fine-tuned per patient. This is the actual test of the "hierarchical, weight-transfer" idea.

Basic architecture: 4 convolution blocks (same idea as Stages 1 & 2) → a bidirectional LSTM that tracks how the signal changes moment-to-moment → an attention layer that blends the important moments into one 128-number "fingerprint" → two small decision heads reading that same fingerprint (detect, then, only if flagged, classify type).



Phase A: training the shared backbone on the detection task (both datasets).



Phase B: subtype fine-tuning via weight transfer, real (left) vs. control (right) — same LOSO protocol as Stage 2.

Results, in three parts:

Phase A — Detection		Value	
Test AUROC		0.824	
Best-F1 threshold		0.65 (sensitivity 0.605, specificity 0.852, F1 0.487)	

Phase B — Subtype (LOSO)		Accuracy	Macro-F1
Real seizure windows		89.3%	0.314
Adversarial control		82.2%	0.301

Phase C — End-to-end cascade (this shared model)	Sensitivity	Conditional acc.	End-to-end
Strict (2 patients, n=4)	100%	100%	100%
Broad (13 patients, n=46)	93.5%	73.8%	67.4%

Pros

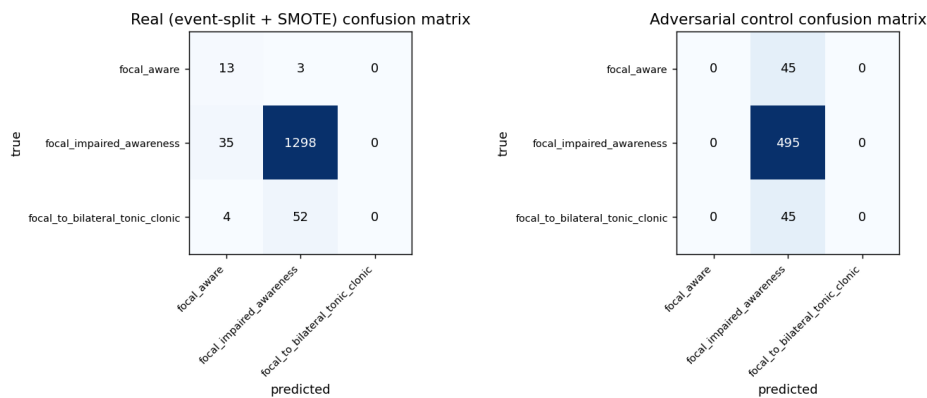
- Architecturally does what was asked: one model, real weight transfer between stages
- Found and fixed a genuine methodology bug (a checkpoint-selection freeze) rather than reporting the inflated result it caused
- Detection performance is competitive with the simpler Stage 1 CNN

Cons

- Does **not** beat the simpler, independently-trained Cascade on any end-to-end number
- Real-vs-control subtype margin (+0.013 macro-F1) is statistically indistinguishable from Stage 2's from-scratch result (+0.015)
- A documented negative result for the weight-transfer hypothesis specifically
- Much slower to train (~80 min) than the plain CNNs (~20 min) for no measurable gain here

What it does: tries to answer "can synthetic data give us better subtype coverage?" honestly. Under strict LOSO, a patient's held-out fold has **zero** real examples of their seizure type to synthesize from — no augmentation trick can fix that. So this deliberately relaxes the split (the same patient can now appear in both train and test, just on different seizure events) specifically to give SMOTE real material to work with. This is a different, weaker claim than "generalizes to a new patient" — and the report says so plainly.

Basic architecture: reuses Phase A's already-trained backbone (frozen — not retrained) purely as a feature extractor. Every window becomes a 128-number "fingerprint"; **SMOTE runs on those fingerprints**, not on raw EEG (interpolating two raw brainwave signals isn't physiologically meaningful — interpolating in the learned feature space is the standard fix). A brand-new, small classifier head is then trained on real + synthetic fingerprints together.



Real windows under the relaxed split + SMOTE (left) vs. the same adversarial control (right).

Results (event-level split, not patient-level LOSO):

	Accuracy	Macro-F1
Real seizure windows	93.3%	0.450
Adversarial control	84.6%	0.306
Real minus control	+8.7 pts	+0.144

For comparison, LOSO's real-minus-control margin was only **+0.013** macro-F1 (Section 4) — this result clears its control by roughly 10× more. **focal_aware** recall went from 0% (LOSO) to **81%** here.

focal_to_bilateral_tonic_clonic still stayed at 0% recall, even with every advantage stacked in its favor — a sign that specific class's problem is the backbone's feature space, not just the sample count.

Pros

- A real, control-verified improvement for one class (focal_aware, 0% → 81% recall)
- Efficient — reuses the existing backbone, trains in under a minute instead of hours
- Transparently documents its own early mistake (an over-aggressive SMOTE ratio that briefly collapsed accuracy to 12%) and the fix

Cons

- Only works by deliberately relaxing patient-independence — not a same-patient-generalization result
- Still 0% recall on the second minority class (focal_to_bilateral_tonic_clonic)
- Ceiling is the frozen backbone's feature space, which was never tuned for subtype separability

Detection numbers are test-set results on held-out patients. Subtype numbers are the "real minus adversarial-control" margin — the one number that actually separates "learned the seizure type" from "recognized the patient."

Model	Detection (test AUROC)	Subtype real acc. / macro-F1	Real – control (macro-F1)	End-to-end (broad)
1. Stage 1 CNN	0.851	—	—	—
2. Stage 2 CNN (LOSO)	—	89.7% / 0.315	+0.015	—
3. Cascade (1+2)	0.851	89.7% / 0.315	+0.015	69.6%
4. Hierarchical CRNN	0.824	89.3% / 0.314	+0.013	67.4%
5. CRNN + SMOTE*	0.824	93.3% / 0.450	+0.144*	—

** Model 5's real-minus-control margin is not directly comparable to the others — it's measured under a relaxed, same-patient-allowed split, not patient-level LOSO. It answers a narrower question ("coverage for patients we have data on") rather than the generalization question the other rows answer.*

Bottom line

Detection works reasonably well and is roughly the same across both architectures tried (~0.82–0.85 AUROC) — the plain CNN and the CRNN are a toss-up here. Subtype classification is fundamentally limited by data, not modeling: two of three seizure types are each backed by exactly one Siena patient, and no architecture change fixes that under a fair, patient-independent test. The weight-transfer idea (Model 4) is architecturally what was asked for, but it did not outperform the far simpler independently-trained Cascade (Model 3) — an honest negative result. The one place real, verified progress was made is Model 5, where relaxing patient-independence (a deliberate, disclosed trade-off) plus embedding-space SMOTE recovered real signal for one of the two previously-invisible seizure types.